

June Resit 2026

Math (Part 1)

Resit Questions (SAT JUNE)

(1) The function $m(x) = c(1.8^x + 1.8^d)$, $c, d \in \mathbb{Z}$, $0 < c < d$. The functions $s(x)$, $t(x)$ are equivalent to $m(x)$. Their are displayed as follows:

I. $s(x) = c \cdot (1.8^x + u)$

II. $t(x) = c \cdot (1.8^x) + v$

Which of the two has the same y-intercept as $m(x)$?

- A) I
- B) II
- C) I & II
- D) Neither

Solution:

The correct choice is D, as:

(y-int of $m(x)$): $m(0) = c \cdot (1 + 1.8^d) = c + c \cdot 1.8^d$

(y-int of $s(x)$): $s(0) = c \cdot (1 + u) = c + c \cdot u$

(y-int of $t(x)$): $t(0) = c \cdot (1) + v = c + v$

For the equality $t(0) = m(0) = s(0)$ to hold, it must be true that:

$u = 1.8^d$ or $c \cdot u = v$ for **every value of u, d, c, v** (which of course is not necessarily the case).

(2) $120 \cdot x^2 - (15p + 21q) \cdot x + p \cdot q = 0$, $p, q \in \mathbb{R}_{>0}$

The sum of solutions of the quadratic is in the form $k \cdot (5 \cdot p + 7 \cdot q)$, find k .

Solution:

Using Vieta's formulas ($S = -\frac{b}{a}$, $P = \frac{c}{a}$), we get:

$$S = -\frac{-(15 \cdot p + 21 \cdot q)}{120} = \frac{5 \cdot p + 7 \cdot q}{40} = \frac{1}{40} \cdot (5 \cdot p + 7 \cdot q),$$

so $k=1/40$.

(3) Function f is defined as $f(x) = -a^x + b$. Let $y=g(x)=f(x)-9$ that has a y-intercept at the point $(0, -\frac{31}{4})$. If the product of a and b is equal to $\frac{225}{4}$, find the value of a and b .

Solution:

$$g(0) = -a^0 + b = 1 + b$$

$$1 + b = -\frac{31}{4} \Rightarrow b = -\frac{31}{4} - 1 = -\frac{35}{4}$$

$$a \cdot b = \frac{225}{4} \Rightarrow a = \frac{\left(\frac{225}{4}\right)}{b} = \frac{\left(\frac{225}{4}\right)}{\left(-\frac{35}{4}\right)} = -\frac{225}{35} = -\frac{45}{7}$$

(4) A transversal line intersects two parallel lines, creating 4 acute and 4 obtuse angles. The measure of one of these angles is equal to $(9x-150)$ - degrees. The sum of four of these angles is k degrees. Which of the following is NOT a possible value for k :

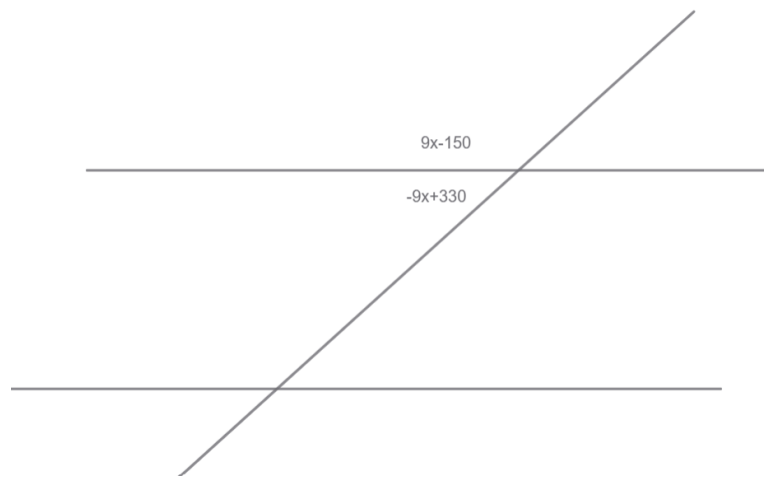
(A) $18x-120$

(B) $-18x+1140$

(C) 360

(D) $-36x+1320$

Solution:



Make a shape to understand the situation, without loss of generality let one of the obtuse angles be $9x-150$. Then the acute will be $180-(9x-150)=330-9x$. Then, for the sum of 4 of these angles, we only have 4 cases:

Case 1: 2 acute + 2 obtuse= 360 degrees

Case 2: 3 acute + 1 obtuse= $3(9x-150)+330-9x=18x-120$

Case 3: 3 acute + 1 obtuse= $9x-150+3(330-9x)= -18x+840$

Case 4: 4 obtuse= $4(330-9x)=1320-36x$

Case 5: 4 acute= $4(9x-150)=36x-600$

So, the correct answer is (B).

(5) The quadratic $g(x) = ax^2 + bx + c$ crosses the x-axis at points $(x_1, 0)$, $(x_2, 0)$ where $x_1 + x_2 = -6$, $x_1 \cdot x_2 = -27$. If $a > 1$ and $a \in \mathbb{Z}$, what could be the value of $b-a$?

(A) 10

(B) 5

(C) 8

(D) 12

Solution:

Solving the system (with Desmos) $x_1+x_2=-6$, $x_1 \cdot x_2=-27$, we get: $x_1 = -9$, $x_2 = 3$ (or the other way around). Now, since $g(x)$ intersects the x- axis at 2 points, it must be true that $D > 0$, so $g(x)=a(x+9)(x-3)=a \cdot x^2 + 6a \cdot x - 27 \cdot a$. Since the two forms of the same quadratic must be equal to each other, $b = 6 \cdot a$, $c = -27 \cdot a$. So $b - a = 5 \cdot a$. Since a is an integer, $b-a$ must be a multiple of 5 (but greater than 5, since $a > 1$). So (A) is the correct choice.

(6) Consider function g such that: $g(x) = k \cdot \sqrt{x + m}$, $(-49, 0)$ is a point of the curve of g and $g(32) < 0$. Which of the following must be true about g ?

(A) $k < m$

(B) $k > m$

(C) $g(0)=49$

(D) $g(0)=-49$

Solution:

Plugging in $x=-49$:

$$g(-49) = 0 \Rightarrow g(-49) = k \cdot \sqrt{m - 49} = 0 \Rightarrow m = 49 \text{ or } k = 0.$$

If $k=0$, then $g(x)=0$ for every value of x in g 's domain, but $g(32)<0$. So $m=49$.

Furthermore:

$$g(32) = k \cdot \sqrt{32 + 49} = 9 \cdot k < 0 \Rightarrow k < 0.$$

So, k is negative while m is positive. So, (A) is correct.

$$(7) \ x(9x - 2k) + m(9x - 2k) - (x + m) = 0$$

k, m : positive constants

The equation above has a solution at $x=50$. Find the value of k .

Solution:

The quadratic equation in its standard form becomes:

$$9x^2 + x \cdot (-2k + 9m - 1) + (-2mk - m) = 0$$

Since $x_0 = 50$ is a **solution**, we get:

$$9 \cdot 50^2 + 50 \cdot (-2k + 9m - 1) + (-2mk - m) = 0 \Rightarrow$$

(solve for k)

$$k = \frac{449m + 22450}{2(50 + m)} = \frac{449(m + 50)}{2(50 + m)} = \frac{449}{2} = 224.5$$

$$(8) \ h(x) = -5(x - u)(x - v)(x - w)$$

For $x \in (u, w)$ (that is: $u < x < w$) $h(x)$ is negative. The graph of $y=h(x)$ in the xy - plane contains the point (m, n) ($m, n \in \mathbb{R}$). If $n=-12$, then which of the following is/ are true?

(I) $m < w$

(II) $u < m < v$

(III) $m > u$

Answers

(A) Only (I).

(B) Only (II).

(C) Only (I), (II).

(D) Only (I), (III).

Solution:

Initially: $f(m) = n = -12 < 0 \Rightarrow m \in (u, w)$

Furthermore:

$$x \in (u, w) \Rightarrow x - u > 0 \text{ and } x - w < 0 \text{ and } f(x) < 0 \Rightarrow$$

$$x - v > 0 \Rightarrow x > v .$$

So:

$$x \in (v, +\infty) \cap (u, w) \Rightarrow x \in (v, w)$$

Correct choice is (D).

(9) The exponential function f is defined by $f(x) = c \cdot d^x$, where c and d are positive constants. If $f(r)=k$, $f(r+1)=k-0.64k$, find the value of d .

Solution:

We have:

$$f(r) = c \cdot d^r \text{ and } f(r+1) = c \cdot d^{r+1} \Rightarrow \frac{f(r+1)}{f(r)} = \frac{c \cdot d^{r+1}}{c \cdot d^r} = d \text{ and } f(r+1) = 0.36 \cdot k, \text{ so}$$
$$\frac{f(r+1)}{f(r)} = \frac{0.36 \cdot k}{k} = \frac{36}{100} \Rightarrow d = 0.36 .$$

(10) The quadratic function $f(x) = x^2 + 3x - 10$ has two solutions r and s , where $r < s$. The quadratic function $g(x) = x^2 + 4x - 21$ has two solutions u and v , where $u < v$. The quadratic function $h(x) = x^2 + 7x + c$ has two solutions $s+u$ and $v+r$ and can be rewritten in the form: $(x + a)(x + b)$. Find the value of c .

Solution:

Solving to find the x -intercepts of $f(x)$ and $h(x)$ we get:

$$f(x) = 0 \Rightarrow x \in \{-5, 2\} \Rightarrow r = -5 \text{ and } s = 2$$

$$h(x) = 0 \Rightarrow x \in \{-7, 3\} \Rightarrow u = -7 \text{ and } v = 3$$

Now, $s + u = -5$ and $r + v = -2$.

So $h(x)$ becomes:

$$h(x) = (x + 5)(x + 2) = x^2 + 7x + 10.$$

So, $c=10$.